

Gamification in software engineering education through Visual Novels

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ABSTRACT

Gamification is not a new term in the field of education. Visual Novels (VN) are text-based and afford interaction with their users. This work discusses the use of VNs as a gamification approach in software engineering education. VNs tell stories and can, therefore, offer a facilitated introduction to complex topics in higher education. Such a topic is the Waterfall model (WM), a process model for software development. It is lectured within a software engineering course at the University of Applied Sciences, Kempten (Germany). This lecture is taught to the students in form of an online video. In order to familiarize students with this topic, we developed and implemented a learning scenario in form of a VN. The aim of this approach is to motivate students, and thus improve their learning success.

CCS CONCEPTS

• **Social and professional topics** → **Software engineering education**; • **Software and its engineering** → *Software creation and management*.

KEYWORDS

software engineering education, visual novel, gamification, learning activation

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1 INTRODUCTION

Gamification uses game elements in non-game contexts [9]. Such a context is software engineering [19], its education [22] and the teaching of it. Many gamification approaches are applied nowadays to improve education in software engineering (e.g. [1, 3]). VNs, however, have not been used for this purpose so far.

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1.1 Definition of terms

VNs have been known for a long time and are widely used in the gaming industry. This video game genre is text-based and offers users many possibilities for interaction. Usually they narrate a story by using textual descriptions, animations, and/or audio content. The viewer of the VN can be included in different ways. Some of these games require just a mouse click to see new texts, scenes, etc. In that case, the story narrates itself (or gets narrated by the games' figures) by the mouse clicks of the viewer. But they can also require their viewer to enter text or choose between different decisions and change the narrated story through these entries. Therefore, VNs can be seen as *interactive comic books*. [6]

An example for a VN can be found in Figure 1, which contains the scenes developed for the VN of this work. How students can be activated by the VN is described in section 2.4. In summary, they have to enter texts, choose between different options, and assign movable objects to their right position.

1.2 Paper context

In software engineering education this high level of interaction can be used to activate and motivate students. The aim of this work is to enhance their learning success. Because VNs tell stories, they provide an opportunity to explain complex topics playfully, and therefore support students during their learning process. This gamification approach can for example ask questions, let students choose between several options, or let them arrange displayed components in the right order. VNs can be used to introduce new topics, or to repeat previous lectures. By not only presenting new educational contents, but also by repeating them in a playful manner, we want to achieve that students actively deal with the course's contents and remember them better.

In order to show how VNs can be used in software engineering education, we developed a story (learning scenario) and implemented it. So far it has not been applied for teaching.

The VN repeats the content of a teaching unit about the WM, which is presented by a YouTube video [11], created especially for the software engineering course at the University of Applied Sciences, Kempten (Germany). The aim of the VN is to activate students by giving them small exercises concerning the contents of the video. It should review and consolidate their knowledge. Because the teaching unit is available as a video, it is possible to jump from the VN to a specific part of the video if the student needs support recalling its content.

At first related work will be discussed and delimited from our work. After that the goals, the VN's use within a software engineering course, the story and its implementation are focused.

1.3 Related work

According to de Sousa Borges et. al. there is a growing interest in the use of gamification approaches in education [7]. Therefore, research has been carried out on visualizing and gamifying educational contents. Jwo uses comics and micro-movies for software engineering education [12]. Barros et. al. make use of comic strips for teaching software engineering [2]. Metraglia and Villa discovered, that (web-)comics seem to motivate weaker students through the use in an engineering graphics course [16]. In an educational context, mostly information comics [5] are used to convey the meaning of selected concepts or contents. This kind of comic, however, only focuses on transferring knowledge. None of these approaches activate students with small exercises and interactions. VNs enable such an activation. According to Lee and Chia, they can be used for a wide range of topics [14].

VNs already have been applied in an educational context by Salazar, Nakajima and Alexandrova [21]. They proposed a methodology for VNs in education, which we applied during this work. They, however, do not apply their methodology on software engineering education. We have not been able to find research work in which VNs have been applied in this educational context. Furthermore, Salazar, Nakajima and Alexandrova concentrated on an implementation for iPads [21]. With our approach, using the Ren'Py engine, VNs can be deployed to the most common desktop and mobile operating systems [20]. To reach a wider range of students, we especially decided to use this engine.

Using Ren'Py to develop VNs is no new approach (e.g. [4]). It has, however, not been used in the context discussed by this work.

2 DIDACTICAL CONCEPT

2.1 Learning goals

The underlying structure of a VN is always a story [6]. According to the methodology of Salazar, Nakajima and Alexandrova, VNs should focus on a specific topic to teach, when they are used in an educational context [21]. Therefore, a story to deal with the WM has been developed as an example for the use of VNs in software engineering education. The procedure described in this work can be applied to other software engineering topics as well. It is important to choose a simple and understandable storyline, to not distract students in their learning process. Because the VN should be used in the software engineering course at the University of Applied Sciences, Kempten, it will be developed in the German language.

The VN shall be accessible by as many students as possible. This is why we chose the Ren'Py engine [20]. So far it has been deployed to a Windows 10 and a Linux Mint 19 operating system. The VN developed in this work should also be available for MacOS users. Future developments can also cover the most commonly used mobile operating systems.

2.2 Course realisation

The executable files, which are the result of the VN created in this work, can be provided to students through an online course on

the learning platform Moodle [17]. All material for the software engineering course is accessible there. The VN will be uploaded in form of executables for the current Windows, Linux and MacOS operating systems.

The VN focuses on consolidating course content, which has already been presented. It can be seen as an "additional material" for an existing teaching unit. We're planning on using it as part of an exercise for the software engineering course at the University of Applied Sciences, Kempten. During the exercise students can download it and work their way through the story. Upcoming questions can either be answered by a teacher or by a functionality, which allows the student to jump to a specific part of the video if he or she couldn't solve one of the exercises.

2.3 Target groups

As a well known process model, the WM is taught in the software engineering course at the University of Applied Sciences, Kempten. According to the study program (for further information please see: [10]) the following courses of studies are targeted as users of the developed VN, within their software engineering courses exercises:

- Business Information Systems (usually in the fourth semester).
- Computer Science (usually in the third semester).
- Computer Science - Game Engineering (usually in the third semester).
- Healthcare Informatics (new course of study; planned for the fourth course semester).

2.4 Developed learning scenario

To create an environment students can identify with, the protagonist of the story is a student who got an internship at the "SwegEd" company (inspired by the name of the YouTube channel the educational video was posted on). The player experiences the VN in 1st person view. During the internship, the protagonist and two other colleagues have to develop a mobile app, which displays the fill level of the developers' coffee pot. This project comes with specific characteristics, which suggest the use of the WM model as process model:

- The project is small.
- The project and its goals are easy to understand.
- The project is self-contained.
- Milestones are easy to define.
- Requirements are defined once and won't change during the project.

According to the video, these characteristics make the use of the WM reasonable [11].

The story consists of ten scenes. For each scene learning goals have been defined (except for scene 0 and scene 1, which are used as an introduction and explanation of the internship project). Table 1 provides an overview of the scenes' contents and their learning goals.

Scene 0 focuses on an introduction for the students. The chief executive officer (CEO) of the SwegEd company congratulates the player for getting the highly coveted internship. Because this scene just introduces the students, it has no learning goal.

Table 1: Learning goals of each scene within the VN

Scene	Content	Learning goal
0	Introduction	None
1	Project explanation	None
2	Choose a process model	Students know when to use which process model
3	Discussion of the project's management methodology	They remember the advantages and disadvantages of WM and SCRUM and arrange them correctly
4	Arrange phases of the WM	Students are able to remember and arrange the phases of the model correctly
5	Requirements phase	They remember User Stories as an option to document requirements and arrange the template components correctly
6	Design phase	Students remember the observer design pattern and arrange its components correctly
7	Implementation phase	They remember how the observer pattern could be implemented and complete a specific code example
8	Verification phase	Students remember different types of tests and assign unit-tests to its belonging type
9	Maintenance phase	Students remember the extension of the WM model by its name (WM with feedback loop)

Likewise, **scene 1** shall not transfer or repeat educational content. It explains the internship project to the player. Both scenes are used to sensitize the students for the following small exercises.

Scene 2 is used to present the above named projects' characteristics. Afterwards, the student is asked whether the WM or SCRUM (e.g. [15, 18]) would fit as a process model for the project. The learning goal of this scene is to revise, under which circumstances the use of the WM makes sense.

Scene 3 discusses why the WM is a reasonable process model in this context. One of the other colleagues argues for, and the other against this model. The player has to assign each argument to the correct employee. The result will be two lists with advantages and disadvantages of the WM. Remembering and assigning them is the learning goal of this scene.

The WM consists of five different phases which are sequentially processed [11]. Their individual names and order of execution should be remembered by the students. This learning goal is queried in **scene 4**. Students have to prove their knowledge by arranging those phases in the correct order.

To achieve a deeper understanding of the WM each phase is regarded individually by the following scenes. Their common learning goal is to repeat and deepen the students knowledge of the stages. Therefore each scene comes with a small exercise, specific for the related phase. These contain content of previous teaching units. This is intended to help students to combine the phases of the model with what they already know, so that they can remember them better. The names of the phases have been taken from the educational video [11] and translated by the authors.

Scene 5 covers the requirements phase. In a different lecture students are taught that requirements can be documented with User Stories. They can be described using a template. The player receives the task to arrange the templates' components in the correct order. [13]

Scene 6 represents the design phase. Within this scene, the interns are working on a notification for the SwegEd developers when the coffee pot has been fully refilled. They are planning to use the observer pattern for this. One of the other colleagues already draw the required classes but can't remember how to arrange them to fit

the pattern. The player is asked to rearrange the classes correctly. Like the User Stories, the observer pattern is discussed in another lesson. To solve this small exercise, the students must apply their knowledge to the given problem.

In **scene 7** the student has to write one line of code, to implement the previously modeled observer pattern. According to the programming language used in the course's script, we chose Java.

The course also discusses several types of software tests (e.g. white-box tests, black-box tests). In **scene 8**, the verification phase, the player has to assign Boundary Value Analysis to its overall type.

Scene 9 represents the last scene and the last phase, the maintenance phase, of the WM. The CEO thanks the interns for their great work. He is satisfied with the chosen process model, but could not remember the name of the model's extension, which allows stepping back to previous phases. According to the educational video, the student's answer must be "waterfall model with feedback loop" [11].

The scenes of the VN described in this section can be seen in Figure 1. They are shown in consecutive order from left to right and top to bottom.

3 IMPLEMENTATION

For reasons of development time, maintenance, and range of deployable systems, the Ren'Py VN engine was chosen. The engine offers a domain specific language (DSL), which interoperates with the Python language. This allows for fast prototyping and implementation. The DSL (called Ren'Py scripts) offers a quick way of implementing common elements in VNs, e.g. dialogues between multiple characters and letting the user choose a dialogue option. Additionally, the Python language can be used for more complex situations, like checking if the user has moved all draggable elements onto their designated target position.

Each scene of the storyline described earlier resides in its own Ren'Py script file. A scene itself is very similar to a callable function. This allows for reordering of scenes outside of the actual scene implementation. Furthermore, scenes can be swapped in and out, without affecting the rest of the storyline.

Since the engine is beginner friendly, regarding programming knowledge, the main work shifts over to dialogue scripting, design, and artwork. The current design consists of four characters in various emotional states, e.g. happy and sad. They represent the company owner, development team members, and a mentor. The player itself is not shown as a displayable figure, and therefore experiences the VN in 1st person view. This is to increase immersion by directly addressing the player [8].

At the moment the only available language for the VN is German.

4 CONCLUSION

In this work a new way of gamified learning in software engineering education has been suggested. VNs are text-based and can provide a high level of interaction with its players. The goal of the VN developed in this work is to revise a teaching unit about the WM. Therefore, an easy understandable storyline was developed. During the gameplay, the player has to solve simple exercises related to the WM and other teaching units.

By using the Ren'Py engine it is possible to deploy the VNs to the most common operating systems, including mobile [20].

The VN should be used as an additional material to a software engineering course at the University of Applied Sciences, Kempten (Germany). We're planning on using it during the courses' exercises to recapitulate the teaching unit about the WM.

Further work should cover a survey on the acceptance level of the VN by students. With a good level of acceptance, other VNs covering more teaching units shall be developed. The project's build process can then be extended to support mobile operating systems as well. Finally, English language support can be added to enable international use.

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Figure 1: Scenes of the VN © Matthias Gensheimer